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Incentivizing digitalization: CEO compensation and digitalization in China's A-share corporate landscape

R.M. Ammar Zahid [a,b], Umer Sahil Maqsood [c], Murtaza Hussain [c], Muhammad Waleed Younas [c,*]

[a] Yunnan Technology and Business University, Kunming, China
[b] Korea University, Seongbuk-gu, Seoul 02481, South Korea
[c] School of Economics and Finance, Xi'an Jiaotong University, Xi'an, 710061, PR China

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ABSTRACT

Executives' compensation structures can strongly affect the ability and readiness of companies to undertake digital transformation (DT) initiatives. This study investigates the intricate dynamics among CEO compensation and DT within Chinese public companies over the period from 2010 to 2023. The analysis identified three principal findings. Firstly, there is a positive association between the CEO paygap and DT, indicating that greater compensation disparities incentivize CEOs to pursue technological advancements. Secondly, managerial shareholding significantly positively moderates this relationship, likely due to the alignment of managerial and shareholder interests. Thirdly, this influence is more pronounced in state-owned enterprises compared to non-state-owned enterprises. This could be due to different governance structures and objectives in state-owned entities. Moreover, government regulations aimed at controlling CEO compensation are found to dampen DT efforts, suggesting that such constraints might hinder digital innovation. Channel analysis reveals agency costs and audit quality as key mediators that help to align the CEOs interests in support of DT. Various methodological approaches, such as instrumental variables (IV) and propensity score matching (PSM), validate the robustness of these results. Findings highlight the need for a balanced approach to CEO compensation and regulation to foster DT. Policymakers should consider the potential unintended consequences of stringent compensation regulations on technological advancement. Companies are encouraged to align managerial incentives with shareholder interests through shareholding schemes to enhance DT efforts. These findings provide a framework for comprehending and managing executive compensation dynamics in the context of digital transformation.

1. Introduction

The pervasive wave of digital transformation (DT) is reshaping industries and businesses globally, driving firms to invest in advanced digital technologies to remain competitive (Maqsood, Wang, & Zahid, 2024). DT has shattered the traditional constraints on economic growth and significantly altered the logic and value creation processes within the business community (Bertani et al., 2021). While the benefits of DT investments (such as firm performance) are well-documented, the substantial costs and management challenges associated with DT present significant hurdles (Kohtamäki et al., 2020). Studies have shown that firms with higher levels of digital maturity tend to outperform their less digitally advanced counterparts (Xing et al., 2023). However, the transition to a digitally mature state is complex and fraught with challenges. One key empirical finding is the critical role of executive leadership in driving DT. Executives, as pivotal strategic decision-makers, play an essential role in navigating their organizations through DT (Wrede et al., 2020). Their commitment and vision are essential for overcoming resistance and aligning organizational resources with digital initiatives. However, short-term interests, risk aversion, and insufficient incentives often deter executives from prioritizing these expensive, long-term digital investments (Jones & Burchman, 2016).

DT is inherently a long-term process whose benefits and returns may not be immediately apparent. This characteristic poses unique challenges for firms, particularly in aligning executive incentives with the prolonged nature of DT benefits. Most of a company's DT projects are

* Corresponding author.
E-mail addresses: amrzahid@gmail.com (R.M.A. Zahid), omersahil@stu.xjtu.edu.cn (U.S. Maqsood), mhussain@stu.xjtu.edu.cn (M. Hussain), waleed@stu.xjtu.edu.cn (M.W. Younas).

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0160-791X/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

suggested and pushed by the executives, and CEOs are crucial in the DT process because they make the most important decisions (Zhu et al., 2022). The management activities largely depend on the compensation and risk level that the corporate executives are involved in. The issue of executive compensation and corporate DT investment essentially involves optimizing the allocation of limited corporate internal resources. There is a strong link between the level of risk and the relationship between executive pay incentives and the input into businesses' DT (Li & Shao, 2023) because risk is a big part of how companies decide to use their resources. This raises the question of how to design compensation packages that not only incentivize immediate performance but also encourage sustained commitment to DT over time. The delayed returns on DT investments (Brynjolfsson & McAfee, 2014; Ha, 2022) make it harder for short-term executive incentives to match up with long-term strategic goals. This highlights an important area that needs more research. From the point of view of corporate governance, the board of directors sets executive pay gaps to make sure that the private interests of executives are in line with the value of the company. This makes it easier to make decisions like DT that increase the value of the company (Khan et al., 2022; Lin et al., 2013). While incentives might drive competitive behavior and potentially foster innovation (Banker et al., 2016), they can also lead to excessive risk-taking and significant pay disparities, which could harm employee morale and organizational cohesion (Lin et al., 2013). Empirical studies in related areas, such as R&D investment show that well-structured compensation packages can encourage innovation and long-term strategic investments (Devers et al., 2008). However, specific studies on the impact of CEO pay gaps on DT are sparse, pointing to a significant research gap. This study seeks to address this gap by investigating the relationship between CEO compensation structures and firms' propensity to engage in DT.

Agency theory and the resource-based view (RBV) of the firm form the theoretical foundation of this research. Agency theory suggests that misaligned incentives may lead managers to prioritize personal interests over shareholder wealth maximization (Jensen & Meckling, 1976). Within the context of DT, this theory suggests that without proper incentives, executives may avoid making significant digital investments due to the associated risks and long-term payoff horizons (Brynjolfsson & McAfee, 2014; Li & Shao, 2023). RBV highlights the importance of leveraging internal capabilities and resources to achieve competitive advantage (Barney, 1991). Researchers view DT as a strategic move that enhances these capabilities by integrating digital technologies. The success of such transformations often hinges on the ability of executives to recognize, acquire, and deploy these resources effectively (Teece et al., 1997). The pursuit of the common goal by managers is more likely to occur if an optimal executive remuneration structure is designed based on the firm's success (Lazear & Rosen, 1981).

Existing research suggests that well-aligned compensation packages can drive innovation and facilitate significant organizational changes (Jensen & Meckling, 1976). Shareholding by managers aligns the interests of executives with those of shareholders. This encourages CEOs to take on DT projects that will create long-term value (Maqsood, Li, et al., 2024; Morck et al., 1988). This alignment mitigates agency problems and encourages strategic decisions that foster innovation and strengthens the impact of the CEO paygap on DT. In the context of China, SOEs and non-SOEs exhibit different dynamics due to their distinct ownership structures and governance models. SOEs, influenced by government policies and political objectives, may have different priorities and incentives compared to non-SOEs, which are driven more by market forces and shareholder expectations (Zahid, Saleem, & Maqsood, 2023). Such differences in incentive structures between state-owned enterprises (SOEs) and non-SOEs influence the CEO motivation towards DT. Additionally, government regulations aimed at reducing the CEO pay gap can enhance employee engagement and collaboration, creating a corporate culture conducive to innovation and technology adoption. In China, the implementation of such regulations in SOEs since 2004 has further shaped the compensation landscape, potentially influencing these firms' propensity to pursue DT initiatives (Maqsood, Li, et al., 2024). Therefore, the interplay between managerial shareholding, the nature of enterprise ownership, and regulatory frameworks collectively impacts the effectiveness of CEO compensation in driving digital transformation. Thus, drawing on agency theory and considering CEO compensation and digitalization strategies, we argue that the relationship between CEO compensation structures and firms' propensity to engage in digital transformation (DT) is significantly moderated by managerial shareholding, the distinction between state-owned enterprises (SOEs) and non-state-owned enterprises (non--SOEs), and government regulations targeting the CEO pay gap.

One major unaddressed puzzle is the specific mechanisms through which CEO compensation impacts DT initiatives. Existing research has predominantly focused on the impact of compensation on general corporate performance or R&D investment (Devers et al., 2008), but there is a paucity of empirical studies examining how these incentives translate into strategic decisions such as digital investments. To address this gap, the current study also investigates that the reduction in agency cost and the role of third-party assurance (Big 4 audit) mediate the relationship between CEO compensation and DT. By aligning CEO interests with shareholder goals, these mechanisms could enhance risk-taking behaviors that are conducive to digital transformation. This finding suggests that agency cost and audit quality act as a potential causal mechanism for the effect of the CEO pay gap on DT. This approach extends the existing literature by linking compensation structures not only to general performance but also to specific strategic outcomes like DT.

Another significant gap in the research is the lack of focus on how contextual factors, such as industry characteristics, access to finances, etc., affect the relationship between CEO pay gaps and digital transformation (DT). It's important to understand these background factors because each industry has its own dynamics, competitive pressures, and regulatory environments that can affect both how CEOs are paid and how digital transformation projects are carried out. Rapid technological advancements in certain regions may prioritize Digital Transformation (DT), thereby influencing the structure of CEO pay that drives and rewards successful DT efforts. Therefore, current research investigates heterogeneous effect regional characteristics and access to the finances in the relationship between CEO pay gaps and DT outcomes. This method might give us more complex information and useful suggestions for changing the pay of executives to encourage digital innovation in various business settings (Gerhart, 2017). In addition, the accelerated shift towards digital solutions and the adoption of digital transformation following the COVID-19 pandemic provide a unique opportunity for examining CEO pay gaps in the context of this rapid DT. A reduced sample analysis, focusing on the post-pandemic period, reveals how CEO compensation strategies adapted to support urgent digital innovation (Gabryelczyk, 2020). This period of forced innovation and adaptation shows different patterns in CEO pay gaps compared to more stable times, highlighting how crisis-induced DT influences executive compensation. This provides a detailed understanding of how external shocks influence internal corporate strategies (Bartulis, 2024; McKinsey&Company, 2021).

To empirically evaluate the hypothesis, the study employs a distinctive secondary data set comprising publicly listed companies in the Chinese stock market for the period spanning 2010 to 2023. China is considered an ideal research setting due to its status as the world's largest and fastest-growing digital economy, offering insights that may generalize to other emerging economies. Second, many Chinese firms have actively pursued DT strategies over time, yielding abundant empirical data to research this phenomenon. Consequently, China offers an opportune real-world laboratory to assess the ramifications of the digital revolution. The study posits that CEO incentives in Chinese firms may positively influence their willingness to undergo DT. Results from panel regressions indicate that CEO pay gaps are significantly positively associated with DT, suggesting that greater compensation disparities

incentivize CEOs to pursue technological advancements. Secondly, managerial shareholding significantly moderates this relationship positively, likely due to the alignment of managerial and shareholder interests. Thirdly, this influence is more pronounced in state-owned enterprises compared to non-state-owned enterprises, potentially due to differing governance structures and objectives. The research also examines the impact of government regulations on CEO compensation, finding that while such regulations initially promote digital advancements, they may subsequently hinder further progress. Agency costs and audit quality mediate the positive impact of CEO pay gaps on DT adoption, according to path analysis. The robustness of these results is validated through various methodological approaches, including instrumental variables and propensity score matching. The findings highlight the need for a balanced approach to CEO compensation and regulation to foster DT. It is suggested that policymakers think about how strict pay rules might slow down technological progress. At the same time, companies are told to use shareholding schemes to align managers' incentives with shareholder interests in order to boost DT efforts.

This study provides a framework for understanding and managing executive compensation dynamics in the context of digital transformation and contributes to both research and practice in a number of ways. First, it contributes new information to the growing body of literature on digital transformation. Existing research has mostly concentrated on the economic and financial outcomes of digital transformation (Kretschmer & Khashabi, 2020; Xing et al., 2023; Zhai et al., 2022) and ESG performance (Ding et al., 2024), whereas the drivers of such a shift have been the subject of relatively few studies. As a result, this study addresses this gap in the literature by investigating the effect of CEOs compensation as a driver for corporations' digital transformation. Second, it extends the CEO compensation literature. While prior research has largely focused on the impact of compensation on general corporate performance or innovation (Banker et al., 2016; Cai et al., 2021), this study uniquely explores how CEOs pay gaps drive DT efforts. Third, the paper provides a theoretical framework about the incentives that motivate corporate leaders in the digital transformation of their companies. It extends agency theory by identifying the underlying mechanisms through which executive compensation affects digital investment decisions (Rashid, 2016). Specifically, it addressed how compensation structures align CEO incentives with long-term digital goals, shedding light on the role of agency costs and risk-taking behavior in digital decision-making processes. Fourth, this research contributes to the literature by elucidating the role of managerial shareholding in aligning executive incentives with strategic goals, particularly in the context of DT. Fifth, the study looked at how environmental factors, like ownership structure (state-owned vs. non-state-owned enterprises), access to finances (high bank concentration vs. low bank concentration), and industrial structure (hi-tech vs. low-tech), affect the link between executive pay and DT. By analyzing the differences between these factors, the study provided valuable insights into how specific organizations and governance structures influence the effectiveness of executive incentives in driving digital change. Sixth, the study advances scholars' understanding of government regulations for the CEO pay gap in the dynamic relationship between CEO compensation and DT. With restrictions on pay gaps, the potential for offering substantial financial incentives linked to DT initiatives is constrained, potentially reducing the motivation for CEOs to aggressively pursue DT. Finally, the paper explores the underlying mechanism for this relationship. A strategically designed CEO pay gap can promote digital transformation by decreasing agency costs and enhancing audit quality. By aligning executive incentives with long-term strategic goals and ensuring high standards of financial reporting, companies can create an environment conducive to successful DT. This alignment not only motivates executives to pursue digital innovation but also builds the trust and confidence needed to support transformative initiatives.

The remainder of the paper is organized as follows. Institutional context and background are presented in Section 2. Literature evaluation and hypothesis development are provided in Section 3. Section 4 presents the empirical setup, including its data sources and empirical model. Results from several empirical tests, such as moderation tests, channel tests, and robustness analyses, are reported in Section 5. Section 6 concludes the study findings.

2. Background and institutional environment

The digital transformation is a recent and rapidly evolving phenomenon, which has led to a scarcity of research due to the challenges in timely and comprehensive analysis (Schwarzmüller et al., 2018). This transformation has resulted in the growth of the global economy and provided unique opportunities for businesses to adopt digital technologies such as big data, the Internet of Things (IoT), and artificial intelligence (AI) to improve their operations and business models. The top five most valuable companies on the S&P 500 index have witnessed significant changes with the emergence of digital and information-rich corporations like Apple, Microsoft, Alphabet, Amazon, and Facebook (Wrede et al., 2020).

The Chinese government is actively promoting DT through policies like the "Made in China 2025″ plan and the "Digital China" initiative. China has emerged as a global leader in fields like AI and 5G, with companies such as Huawei, Baidu, and Tencent investing heavily in research and development. Changing consumer needs, the rise of e-commerce and mobile payments, and the use of big data are driving digital transformation in China.

The growth of China's economy, financial markets, and digitalization highlights the importance of well-established corporate governance processes to protect investor interests. Despite the criticisms, Chinese executives and businesses are showing a readiness to strengthen their corporate governance practices (Clarke, 2003). Fig. 1 shows a rising trend of DT in overall and SOEs vs. non-SOEs of China over time. Particularly, upward slopes significantly shifted in 2005 and 2020. The China Listed Company Governance Index (CCGINK) has shown continuous progress over the years. Public companies have adopted executive pay plans and disclosed executive remuneration details (Zhang et al., 2018).

Fragmentation in the existing literature on DT makes it challenging to understand its wide-ranging effects and identify major shifts (Ha, 2022). Its adoption significantly transformed production techniques and organizational structures inside firms, garnering considerable interest in the field of economics (Jiang & Li, 2024; Nambisan et al., 2017). Research shows that DT can enhance a company's enterprise value, competitive performance (Zhai et al., 2022), productivity (Kretschmer & Khashabi, 2020), innovation (Xing et al., 2023), and ESG performance (Ding et al., 2024). Studies also evidence the use of digital technology in corporate value generation, indicating that such technologies optimize internal corporate processes and augment organizational efficiency performance (Maqsood, Wang, & Zahid, 2024). Others acknowledge that digital transformation involves more than technological adoption; it also requires strategic management, human resources, and cultural support (AlNuaimi et al., 2022). However, businesses struggle with DT due to substantial ongoing expenditures and resource constraints (Moktadir et al., 2018). To maintain or improve their competitive position, companies must undergo deep, systematic changes. CEOs may have less incentive to support DT investments due to their long-term nature, compared to other forms of long-term capital investment.

3. Hypothesis development

The existing literature shows that people within a company drive its digital transformation, with executives playing a crucial role (Benitez et al., 2022). Senior managers have a significant impact on company strategy and performance (Certo et al., 2006). The level of managerial discretion, or the influence of senior managers on outcomes, is

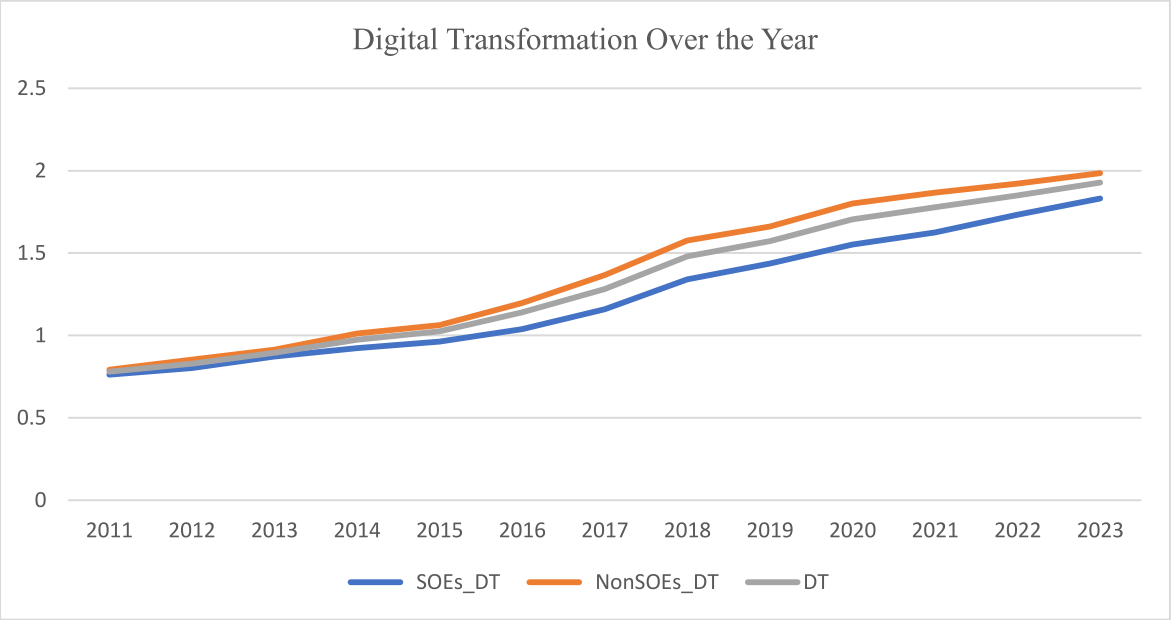


Fig. 1. Journey of digital transformation (DT) in the overall firm and DT by the type of firms state-owned enterprises (SOEs) vs non-State-owned enterprises (non-SOEs).

important to investigate (Wangrow et al., 2015). CEOs, in particular, face unique challenges in maintaining their company's competitiveness during the digital transition (Hess et al., 2016).

A firm's digital ability depends on its resources, including technology, expertise, and capital (Khin & Ho, 2018). The uneven growth of China's digital economy leaves some firms with limited digital foundations and a scarcity of essential tools (Jackson & Dunn-Jensen, 2021; Kane et al., 2017). Excessive costs and limited access to capital pose barriers to DT (Raj et al., 2020). The decision-making and actions of top managers, especially CEOs, have significant repercussions in this context (Quigley & Hambrick, 2015). However, the academic debate has largely ignored the motivations of senior managers toward the digital revolution. According to previous research by Cortellazzo et al. (2019) and Wrede et al. (2020), the digital transformation cannot occur without the backing of upper management. But there is still a lack of clarity regarding CEOs motivation in corporate digital transformation efforts, despite the practitioner literature highlighting the importance of CEOs compensation in IT investments (Ravichandran & Zhao, 2018).

The study's theoretical framework is based on the concept of behavioral forces. Specifically, agency theory suggests that executives and stockholders may have distinct agendas and approaches to taking on risk (Jensen & Meckling, 1976). Executives may have more instrumental objectives centered on their personal and professional advancement, in contrast to shareholders who strive to maximize the returns on their investments and the overall worth of the business. Therefore, well-designed executive compensation structures can align management behavior with shareholder interests (Jensen & Murphy, 1990). Furthermore, information asymmetry between ownership and management may lead to managerial opportunism (Jensen & Meckling, 1976). Previous research has demonstrated that CEOs' risk propensity influences their support for digital investments (Cortellazzo et al., 2019; Wrede et al., 2020). Agency difficulties, such as idleness and excessive spending, arise due to the challenges of monitoring managers' behavior and resource allocation (Hirshleifer & Suh, 1992). Balancing limited resources among conflicting demands is a challenge for CEOs (Lambert, 1986). The resource-based theory asserts that scarce resources, which are inimitable and irreplaceable, constitute the foundation of competitive advantage for businesses (Grant, 1991). Digital transformation activities are value-generating endeavors that enable firms to acquire resources. Augmenting DT investment may enhance the current production procedure, lower manufacturing expenses, and elevate corporate performance and firm worth (Chwiłkowska-Kubala et al., 2023; Xing et al., 2023). Nonetheless, the prolonged cycle and high cost of DT operations introduce significant uncertainty and risk. With the failure of DT efforts, the business risks losing market share and disrupting the financing cycle. The company may experience financial difficulties or declare bankruptcy. Consequently, in situations of knowledge asymmetry, managers will reduce DT spending, adversely affecting the firm's value. To mitigate moral hazard and enhance managerial motivation for DT initiatives, corporations are inclined to provide substantial compensation to management.

The need for higher executive pay arises from the fact that companies want to recruit and retain highly skilled leaders who can spot and capitalize on the chances to increase their firm's worth (Certo et al., 2006; Jensen & Murphy, 1990). A greater pay disparity between executives and non-executives and better business performance in China (Banker et al., 2016) are all consistent with a higher executive pay premium. Executive remuneration based on observable signals, such as an executive's prior performance, is proposed by managerial talent theory as a solution to the adverse selection issue emerging from ambiguity regarding an executive's abilities. Executives who have consistently delivered results are valued more highly and compensated accordingly. Therefore, managers who receive higher compensation than their peers are more likely to possess skills, leading to increased firm value for their businesses. Conversely, the theory of relative exploitation posits that an escalation in the executive pay disparity diminishes healthy competition and erodes the sense of collaboration and accountability within the organization, consequently inciting feelings of inequality and organizational deprivation among non-core executives (Lin & Lu, 2009). Thus, increased CEO pay disparity prompts non-core executives to use riskier strategies to address the perceived injustice (Kini & Williams, 2012). Non-core executives significantly influence a company's digital transformation processes. When confronted with a significant pay disparity, non-core executives are prone to perceiving their compensation as inequitable, leading to a decline in their work efforts (Wang et al., 2015). This, in turn, diminishes their support for digital transformation-related policies and their enthusiasm for engaging in digital transformation practices and initiatives.

The digital transition involves both risks and rewards, with CEOs facing the challenge of allocating resources wisely (Brynjolfsson & McAfee, 2014). Investments in digital technologies often don't show results for companies' bottom lines until at least three to five years after they've been made (Brynjolfsson & McAfee, 2014); therefore, patience is required when making large-scale investments in digital technology. The negative cash flow may tempt CEOs to forego digitalization in the near term. Finally, there is a high failure rate among digital transformation projects. CEOs have the unenviable job of dividing limited resources among conflicting demands. A common response to the greater uncertainty surrounding investments in digital technology is to reduce spending in certain areas (Hirshleifer & Suh, 1992; Lambert, 1986). While such avoidance behaviors may help in the short term, they may be counterproductive in the long run in regard to developing digital skills that might provide a strategic advantage. Institutional pressure from stakeholders may also influence digital investment decisions; CEOs may feel pressured to engage in digital transformation, regardless of whether such expenditures align with the company's requirements, as they fear losing legitimacy with the firm's stakeholders if they do not make digital technology investments. CEOs play a crucial role in setting the company's digital strategy and resource allocation (Ravichandran & Zhao, 2018).

The executive pay gap serves as a significant indication of a firm's compensation structure and is associated with the critical strategic choices made by key executives (Banker et al., 2016). As the disparity in CEO compensation expands, stakeholder scrutiny of the business and its leaders intensifies (Faleye et al., 2013). In light of the pressure from these stakeholders, the highly compensated core executives perceive the compensation disparity as validation of their performance and capabilities, thereby motivating them to diligently execute their responsibilities and make decisions that optimize the firm's interests (Khan et al., 2022). Investments in digital transformation are long-term, ongoing, and capital-intensive decisions made by CEOs. Firms in China are experiencing a tremendous deal of uncertainty as the country continues to explore the possibilities of digitization, which is still in its early stages. The ability of the leadership to make sound strategic choices is now crucial to the achievement of digital transformation goals. Moreover, in companies with substantial pay disparities, key executives exert increased influence and play a more pivotal role in critical corporate choices (Wang et al., 2015), including the formulation, implementation, and upkeep of digital technologies. When structured strategically, this pay gap can align CEO incentives with the long-term digital goals of the company, thereby promoting DT. A substantial portion of a CEO's compensation tied to performance metrics related to digital initiatives can motivate the CEO to prioritize and drive DT, ensuring that the organization remains competitive and continues to create value over the long term (Jensen & Meckling, 1976). Incentive alignment through the pay gap encourages CEOs to take calculated risks necessary for successful DT. The potential for higher rewards can lead CEOs to invest in innovative technologies and processes, which might initially appear risky but hold the potential for significant business transformation and returns (Murphy, 1999). This approach ensures that CEOs are not only motivated by immediate financial gains but are also committed to the strategic digital vision of the company. Hence, the CEO pay gap, when strategically designed, serves as a powerful lever for promoting digital transformation. By aligning CEO incentives with digital goals, encouraging risk-taking, and ensuring resource allocation focused on innovation, companies can effectively drive their DT agendas. Consequently, we propose that in organizations with significant pay disparities, senior executives wield increased authority and incentive to DT.

H1. The pay gap between CEOs and other employees positively influences the digital transformation of a company.

3.1. Moderating effects of managerial shareholding

Managerial shareholding, referring to the ownership of company shares by managers or executives, plays a significant role in the CEO's incentive alignment for the DT initiatives. The primary agency problem between managers and shareholders is the possible misuse of free cash flow by self-interested managers, who may prioritize personal gain or engage in empire-building to the detriment of shareholders (Zahid, Khan, & Kaleem, 2023). Alignment theory posits that managerial ownership functions as a governance mechanism (Jensen & Meckling, 1976), with its augmentation facilitating the alignment of interests between managers and shareholders. Increased equity holdings by managers incentivize them to act as owners, thereby making decisions that more transparently reflect shareholders' interests (Jensen et al., 1992; Jensen & Meckling, 1976; Shu & Thomas, 2019). Conversely, entrenchment theory posits that managers may secure their positions to the detriment of shareholders and prioritize personal interests when their ownership stakes are substantial (Morck et al., 1988). Increased managerial ownership confers greater power and influence, potentially leading to a disregard for the risk of replacement or market discipline when they possess adequate authority to safeguard their positions or dominate the board (Lafond & Roychowdhury, 2008; Morck et al., 1988; Shu & Thomas, 2019). Nonetheless, corroborative evidence for this is mostly seen in developed markets, where companies often exhibit more equitable shareholder ownership arrangements and enhanced protections for minority shareholders. In contrast to Western markets, most publicly traded companies in the Chinese stock markets include a controlling shareholder who possesses the authority to choose executive compensation, appointments, and terminations (Conyon & He, 2011; Firth et al., 2006). Thus, managerial ownership in China provides a unique environment to test its effect on CEO incentives.

The foregoing discussion provides support for the idea that the significant number of shares held by the executives in the company can act as a moderating factor against the agency problem regarding investment in corporate digital transformation. The rationale behind this is that managerial shareholding aligns the personal financial interests of managers with those of the shareholders (Jensen & Meckling, 1976), thereby incentivizing managers to take actions that enhance shareholders' wealth. Thus, managerial shareholding plays a crucial role in influencing corporate investment decisions and mitigating agency issues. There are two schools of thought on the impact of management ownership on risk-taking and firm value in the literature (Chen et al., 1993). However, there is consensus that managerial abilities may help reduce the agency problem (Rashid, 2016). We propose that CEOs' actions in guiding digital transformation at their companies depend on elements outside financial incentives, such as management ownership. Hence, managerial shareholdings motivate chief executive officers to put more money into the digitalization of their companies. This leads us to the following hypothesis.

H2. Managerial shareholding positively moderates the association between CEO's pay gap and digital transformation.

3.2. Pay gap and digital transformation—ownership paradox (SOEs vs. non-SOEs)

Political ownership is a widespread mechanism in China's national economy (Jin et al., 2022), and nearly half of China's publicly traded enterprises are controlled by the state, giving the Chinese government a considerable voice in company affairs (Banker et al., 2016). Executives of state-owned firms in China function as "quasi-officials" rather than as professional managers. A significant number of CEOs are or have been government officials. Some executives expect to return to government after gaining managerial experience. Moreover, under governmental administrative interference, profit maximization is not the only objective of state-owned firms. They must endure policy obligations and

various aims, such as ensuring social stability and fostering employment (Yin & Du, 2021a,b).

SOEs in China often have dual objectives, balancing commercial success with socio-economic goals such as employment stability and national strategic interests (Zahid et al., 2024). This dual mandate can affect CEO compensation structures, which are frequently influenced by government policies and regulations that may emphasize stability and risk aversion over aggressive digital innovation (Fan et al., 2007). Consequently, the governance framework within SOEs tends to be more bureaucratic, potentially impacting the speed and nature of decision-making processes related to DT initiatives. In contrast, non-SOEs in China are primarily driven by profitability and shareholder value, which incentivizes a more aggressive pursuit of DT to maintain competitiveness and market leadership (Chen et al., 2011). CEO compensation in these firms is often closely tied to performance metrics, including innovation and digital initiatives, which can motivate executives to actively engage in DT efforts.

The risk appetite for innovation also differs between these types of enterprises. SOEs, due to their public accountability and potential political repercussions from failures, may exhibit a lower risk tolerance, leading to a more cautious approach to DT that focuses on incremental rather than radical innovation (Xu, 2011). Conversely, non-SOEs, driven by competitive pressures and the need for market differentiation, typically display a higher risk appetite, resulting in a more proactive and aggressive approach to adopting and integrating digital technologies (Liu et al., 2013).

Regulatory environments further differentiate the DT approaches of SOEs and non-SOEs in China. Closely aligned with government objectives, SOEs may encounter stringent regulatory environments that can both stimulate and impede DT efforts. While government policies can mandate certain digital initiatives, they may also impose constraints that limit flexibility (Peng et al., 2016). Non-SOEs, on the other hand, often operate under less restrictive regulatory conditions, allowing for greater experimentation and rapid implementation of digital technologies and thus enabling them to adapt swiftly to market changes and technological advancements (Peng et al., 2016).

It has also been discovered that the beneficial effect of CEOs incentives on company performance is less for companies that have a higher percentage of state ownership (Banker et al., 2016). One possible explanation for this is the unusual nature of the executive remuneration packages offered by SOEs. One's political standing has a significant impact on the political benefits that are included in the entire remuneration package for CEOs in SOEs. High-ranking officials at state-owned enterprises (SOEs) may be eligible for lifetime benefits (not included in the pay plan), such as generous pensions, housing subsidies, free or low-cost medical care, and job stability (Kato & Long, 2006). Recent years have seen a tightening of the "pay cap" regulation for SOE managers' salaries, bringing the average executive wage down to around 7 or 8 times the salary of the ordinary SOE worker (Zhang et al., 2018). As a result, the demand for higher remuneration among SOE leaders has intensified.

Therefore, while both SOEs and non-SOEs in China are engaged in digital transformation, their approaches are shaped by differing motivations, incentives, and constraints. SOEs may adopt a more conservative stance toward DT due to regulatory and socio-political considerations, whereas non-SOEs pursue DT more aggressively, driven by market competition and performance-based incentives. Understanding these contextual differences is crucial for developing strategies that align the CEOs incentives with DT. Consequently, we anticipate that the correlation between the CEO compensation disparity and DT is more significant in non-state-owned enterprises.

H3. The association between CEOs' pay gap and digital transformation is significantly negatively moderated by state ownership.

4. Research methodology

4.1. Samples and data

The study sample is comprised of Shanghai and Shenzhen A-share listed firms from 2010 to 2023. The sample begins in 2010, marking the year when Chinese A-share listed firms started integrating digitalization into their business strategies (Wu & Li, 2024). Thus, the data reflects the early stages of this transformation while also capturing the more recent period of rapid digitalization that has been a key driver of economic and business growth in China. Fig. 2 provides a detailed flow of research methodology. The data on digital transformation and corporate governance characteristics was sourced from the China Stock Market and Accounting Research Database (CSMAR) and the Chinese Research Data Service Platform (CNRDS). After screening financial institutions and "ST" (special treatment) firms with insufficient data, the final sample consists of 23,746 firm-year data. Furthermore, all continuous variables are winsorized at the 1 % and 99 % levels to limit the influence of outliers. We employ Python for data mining and textual analysis during data collection and Stata 15 for data processing.

4.2. Key variables measurement

This study's independent variable is the CEOs pay gap, determined by the salary gap between the CEOs and other corporate executives. Specifically, the CEOs pay gap is computed by taking the natural logarithm of the CEO's compensation subtracted from the sum of all executives' compensation (PayGap1), in accordance with earlier Chinese research (Banker et al., 2016). Secondly, the natural logarithm of the pay gap between the total salary of CEOs (including bonuses, etc.) and that of the executives (PayGap2) is an alternative proxy.

$$\text{PayGap1} = \text{nLog}\,(\text{CEO Pay} - \text{Executives Pay}) \tag{1}$$

$$\textit{PayGap2} = \textit{nLog}\,(\textit{Total CEO Pay} - \textit{Executives Pay}) \tag{2}$$

Finally, the natural logarithm of the pay gap between the total salary of CEOs and the average pay of the executives (PayGap3) is used as the instrument variable for 2SLS. The dependent variable of this study is digital transformation (DT). As was done in earlier research by Maqsood, Wang, and Zahid (2024), a DT index was made by using a dictionary-based text analysis of the annual reports of publicly traded companies. The steps in the process are finding words that are related to digital transformation, making a dictionary with all possible words (a Chinese wordlist with digital keywords that makes digital keywords logical and complete), using a machine learning algorithm to find out how often these words appear in the annual reports of sample companies, and using the dictionary-based method to figure out the level of digital transformation. This is done by normalizing the frequency counts by the length of a certain portion of the annual report, specifically the management discussion and analysis (MD&A).

This analysis starts by categorizing policy papers on digital transformation and extracting key terms. Papers like "China's 14th Five-Year Plan," "Made in China 2025," "Digital Economy and its Core Industry Statistical Classification 2021," and "Implementation Plan for Cultivating New Economic Development" are included. The terms cover digital technology (blockchain, virtual reality, big data, AI, cloud computing) and business model change. Next, a deep learning model was used to find nine keywords that had similar meanings to the basic terms in the reports. These keywords were then used to build a full digital dictionary by adding to the basic terms.

Following Brown and Tucker (2011), the focus was on the management, discussion, and analysis (MD&A) sections in annual reports, resulting in the collection of 37,266 MD&A documents. We trained the RoBERTa-wwm-ext model on all collected MD&A, adhering to the methodology of Cai et al. (2021). To create the digital lexicon, the ongoing results from the deep learning model were utilized. Keywords

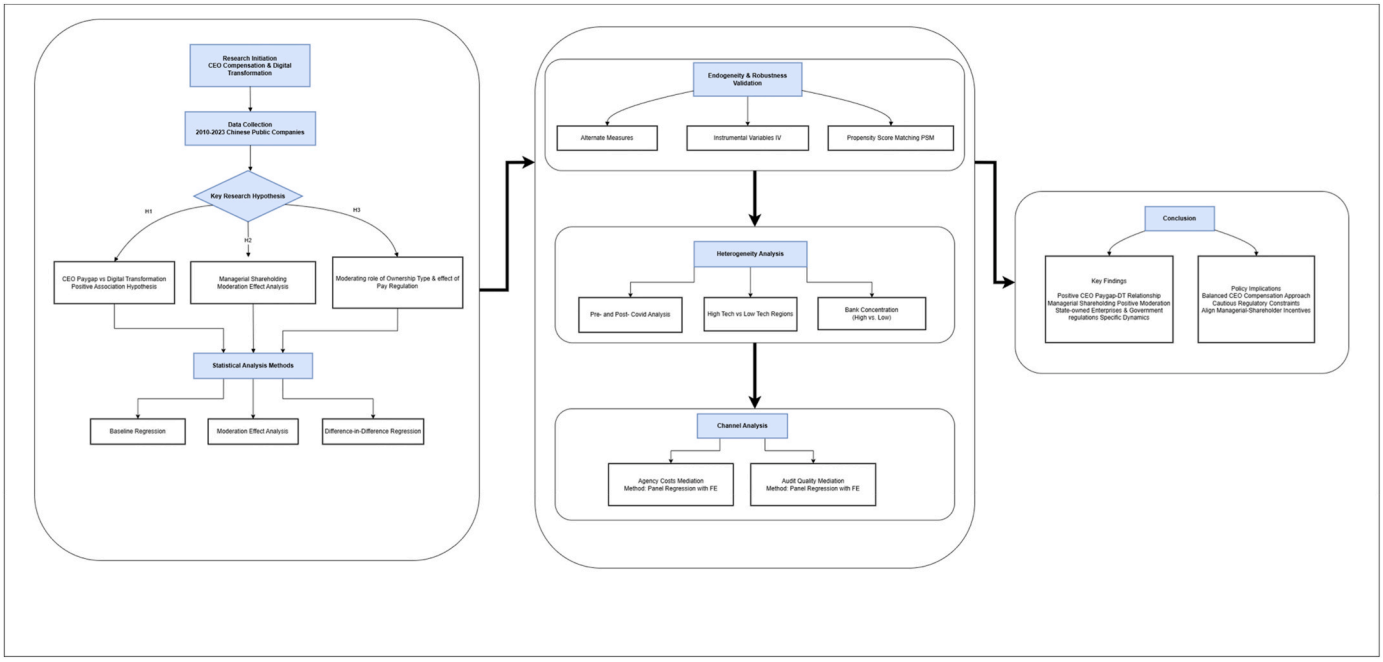


Fig. 2. Shows the PSM covariates balance test.

with a similarity score above 0.80 to the core phrases were retained, while irrelevant keywords were discarded. This process merged basic and extended phrases, resulting in an 86-keyword Chinese digital lexicon. The total word frequency of all digital keywords was transformed using the natural logarithm and served as the surrogate variable "Digital" for evaluating enterprises' digital operations.

4.3. Control variables measurement

Several control factors relating to company digital transformation are used, based on earlier research (Ravichandran & Zhao, 2018; Wu et al., 2022; Zhai et al., 2022). Specifically, the study includes the following control variables: Debt-asset ratio (*LEV*), natural logarithm of firm total assets *(Size)*, rate of return on assets (*ROA*), big four auditors (Big4 audit), and number of employees (N_employees). Additionally, board-level characteristics are controlled for, including the total members on the board (Bosize), Independent director (ind_Dir), and CEO duality (Duality). Year and industry are also controlled for. The variable definitions are shown in Table A1 (appendix).

4.4. Research model

In order to evaluate the impact of tournament incentives on digital transformation, the following regression model has been formulated:

$$DT_{it} = \alpha + \beta_1 PayGap_{it} + \beta_n control_{it} + year_t + industry_i + \varepsilon_{it} \quad (3)$$

Where; DT_{it} is dependent variable, refer to Digital Transformation calculated in firm *i*, year *t; PayGap*$_{it}$ is CEOs pay gap measured as PayGap1 and PayGap2, alternatively. *Control* are a series of firm level and CG control variables. Industry and year fixed effects were also included.

Table 1
Summary statistics.

	Panel A					Panel B		Panel C	
	Total sample *(N* = 32,588*)*					SOEs sample *(N = 13135)*		Non-SOEs sample *(N = 19453)*	
	Mean	max	min	Median	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
DT	1.409	6.282	0.693	0.693	1.081	1.189	0.886	1.542	1.164
DT2	8.735	534.000	0.000	1.000	25.384	5.043	16.693	10.982	29.234
PayGap1	13.863	17.888	5.519	13.907	1.114	13.944	1.127	13.707	1.076
PayGap2	14.414	18.119	7.601	14.456	0.973	14.463	0.996	14.363	0.946
PayGap3	15.116	18.736	6.571	15.124	0.834	15.061	0.849	15.050	0.804
Size	22.110	26.452	18.266	21.939	1.326	22.264	1.397	22.060	1.314
Lev	0.440	3.295	0.026	0.427	0.233	0.489	0.243	0.420	0.227
Age	2.215	3.401	0.693	2.398	0.763	2.573	0.657	2.107	0.714
ROA	0.034	0.277	−0.672	0.037	0.077	0.030	0.074	0.033	0.079
CR	0.163	0.841	0.002	0.126	0.130	0.151	0.120	0.168	0.136
PPE	0.217	0.784	0.001	0.183	0.165	0.233	0.182	0.211	0.156
BoSize	8.662	18.000	1.000	9.000	1.748	9.288	1.889	8.237	1.495
IndDir	37.374	100.000	14.290	33.330	5.578	36.915	5.795	37.674	5.403
N_employee	7.580	13.223	0.693	7.533	1.354	7.989	1.441	7.305	1.209
FCEO	0.029	1.000	0.000	0.000	0.169	0.031	0.173	0.028	0.165
Duality	0.266	1.000	0.000	0.000	0.442	0.095	0.294	0.378	0.485

Note: Where DT = Digital transformation; DT2 = Digital transformation 2; PayGap1 = CEOs Pay Gap1; PayGap2; CEOs Pay Gap2; PayGap3; CEOs Pay Gap3; Size = Firm total size; Lev = Firm leverage; Age = Firm age; ROA = Return on assets; CR = Cash ratio; PPE = Propriety plant and equipment; BoSize = Board size; IndDir = Independent director; N_employee = Employee size; FCEO = Female CEO; and Duality = CEO duality.

The intercept and the random error term are denoted by the letters α and ε. Table 1 defines all variables.

5. Empirical results

5.1. Descriptive statistics and correlation matrix

In Table 1, Panel A reveals the overall sample descriptive statistics, while Panels B & C present the means and standard deviations of subsamples, i.e., SOEs and non-SOEs. The mean values of CEOs' pay gaps, i.e., PayGap1 and PayGap2, are 13.863 and 14.414, respectively, and their associated standard deviations are 1.114 and 0.973, demonstrating that there is substantial variation among firms in CEO compensation. The mean and standard deviation of digital transformation (DT) are 1.409 and 1.081, respectively, indicating an overall lower value that varies between different companies, and all control variable statistics are compatible with the existing body of knowledge. Furthermore, the mean pay gaps are quite similar in both subsamples, i.e., SOEs and non-SOEs. However, the mean value of DT in non-SOEs is relatively high when compared to SOEs, which indicates the high significant value of the adoption of digital transformation.

Table 2 provides the Pearsons' correlation matrix, revealing a significant positive correlation between the DT and CEOs' pay gap. Additionally, the variance inflation factors (VIF) for all coefficients range from 1.002 to 1.458. These values indicate that multicollinearity is not a significant concern in this analysis. Therefore, we can confidently interpret the results, indicating a robust relationship between the DT and the pay gap among CEOs.

5.2. Baseline regression results

Table 3 displays the baseline regression estimations. CEOs paygap was regressed on firm digital transformation with the set of control variables as described in equation (3). Table 3, columns 1 & 2, report the results of OLS regression, and columns 3 & 4 report the results with fixed effect regression with industry and year fixed effects for two different measures of CEOs, pay gap, i.e., PayGap1 and PayGap2, alternatively.

The results indicate a positive association between the CEO's pay gap and DT. Specifically, the coefficients of the Paygap1 and Paygap2 variables in columns 2–5 of both OLS and fixed effect regression estimates with industry-year fixed effect models are significantly positive, i.e., 0.096*** (S.E. = 0.006), 0.024*** (S.E. = 0.005), 0.119*** (S.E. = 0.007), and 0.030*** (S.E. = 0.007), respectively. The results demonstrate that the pay gap between CEOs and other employees enhances the degree of digital transformation of companies, thereby supporting H1. The results are consistent with agency theory and managerial incentives because the pay gap in China is likely to be the motive for CEOs to adopt DT by reducing the agency costs. The calculated coefficient of control variables is according to expectation and prior studies (Jackson & Dunn-Jensen, 2021; Li & Shao, 2023; Zhu et al., 2022). For example, the coefficients of Size, N_employee, and Ind_directors exhibit positive and statistically significant values at the 1 % level. This aligns with the common understanding that agency conflict tends to be lower in larger, established firms with independent directors. Further, the coefficients of Lev, Age, CR, PPE, and BoSize are significantly negative, representing that a large board, higher dependency on debts, age, and lower cash levels hinder the digital transformation of companies.

5.3. The moderating effect of managerial shareholding in CEOs pay gap and DT nexus

Furthermore, the moderating effect of managerial shareholding (MSH) in the CEOs paygap and DT relationship was estimated. To explore this, two interaction terms (PayGap1 × MSH) and (PayGap2 × MSH) were estimated in the model as explained in the following equation (4) under hypothesis 2 (H2).

Table 2
Pearson correlation.

	DT	PayGap1	PayGap2	Size	Lev	Age	ROA	CR	PPE	BoSize	IndDir	N_employee	FCEO	Duality	Variance inflation factor	
															VIF	1/VIF
DT	1															
PayGap1	0.191***	1													1.273	0.785
PayGap2	0.221***	0.898***	1												1.316	0.76
Size	−0.0410***	0.0460***	0.0395***	1											1.458	0.686
Lev	−0.0723***	0.0213***	0.00926	0.334***	1										1.358	0.736
Age	−0.146***	−0.00225	−0.0218***	0.372***	0.344***	1									1.221	0.819
ROA	0.0226***	0.0152**	0.0162**	0.0438***	−0.348***	−0.169***	1								1.357	0.737
CR	0.0360***	0.00330	0.0127*	−0.245***	−0.378***	−0.284***	0.243***	1							1.114	0.898
PPE	−0.0467***	−0.00721	−0.00780	0.0886***	0.0922***	0.0970***	−0.0680***	−0.317***	1						1.423	0.703
BoSize	−0.0944***	0.177***	0.185***	0.0410***	0.0707***	0.139***	−0.00295	−0.0372***	0.00187	1					1.29	0.775
IndDir	0.0703***	−0.0138*	−0.0218***	−0.0116*	−0.0144*	−0.0295***	−0.00107	0.0100	−0.0111	−0.448***	1				1.332	0.751
N_employee	0.0671***	0.380***	0.415***	0.0542***	0.0993***	0.167***	−0.0100	−0.0416***	0.0325***	0.245***	−0.000169	1			1.002	0.998
FCEO	−0.0125*	−0.00317	−0.00199	−0.00856	−0.0201***	−0.00542	0.0125*	0.0234***	−0.0322***	0.00123	−0.00107	−0.00786	1		1.068	0.936
Duality	0.109***	−0.0209***	−0.0533***	−0.0238***	−0.0804***	−0.173***	0.0160**	0.0226***	−0.0290***	−0.182***	0.105***	−0.127***	−0.00982	1		

Note: Where DT = Digital transformation; DT2 = Digital transformation 2; PayGap1 = CEOs Pay Gap1; PayGap2; CEOs Pay Gap2; PayGap3; CEOs Pay Gap3; Size = Firm total size; Lev = Firm leverage; Age = Firm age; ROA = Return on assets; CR = Cash ratio; PPE = Propriety plant and equipment; BoSize = Board size; IndDir = Independent director; N_employee = Employee size; FCEO = Female CEO; and Duality = CEO duality.
*p < 0.05, **p < 0.01, ***p < 0.001.

Table 3
Baseline result regression results.

DV = DT	OLS regression		Fixed effect regression	
	(1)	(2)	(3)	(4)
PayGap1	0.096***		0.024***	
	(0.006)		(0.005)	
PayGap2		0.119***		0.030***
		(0.007)		(0.007)
Size	0.006	0.003	0.013***	0.018***
	(0.005)	(0.005)	(0.005)	(0.005)
Lev	−0.108***	−0.096***	0.024	0.026
	(0.030)	(0.029)	(0.023)	(0.022)
Age	−0.100***	−0.097***	−0.006	−0.019*
	(0.010)	(0.009)	(0.010)	(0.010)
ROA	0.038	0.054	0.055	0.045
	(0.084)	(0.080)	(0.055)	(0.052)
CR	−0.146***	−0.151***	−0.053	−0.062*
	(0.052)	(0.050)	(0.038)	(0.037)
PPE	−0.250***	−0.234***	−0.022	−0.009
	(0.044)	(0.043)	(0.036)	(0.035)
BoSize	−0.027***	−0.031***	0.024***	0.025***
	(0.004)	(0.004)	(0.004)	(0.004)
IndDir	0.003**	0.004***	−0.001	−0.000
	(0.001)	(0.001)	(0.001)	(0.001)
N_employee	0.044***	0.040***	0.138***	0.120***
	(0.005)	(0.005)	(0.007)	(0.006)
FCEO	−0.064*	−0.058*	−0.010	−0.023
	(0.034)	(0.033)	(0.026)	(0.025)
Duality	0.139***	0.141***	−0.006	−0.002
	(0.013)	(0.013)	(0.012)	(0.012)
Constant	−0.654***	−0.913***	−0.919***	−0.949***
	(0.149)	(0.150)	(0.147)	(0.148)
Industry FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Observations	32332	32332	32332	32332
R-squared	0.170	0.174	0.308	0.307

Notes: Standard errors are in parenthesis ***p < 0.01, **p < 0.05, *p < 0.1. Variables definitions are given in Table A1(Appendix). Estimated Model is explained in section 4.5, equation (3).

$$DT_{it} = \alpha + \beta_1 PayGap_{it} + \beta_2 PayGap \times MSH_{it} + \beta_n control_{it} + year_t + industry_i + \varepsilon_{it} \quad (4)$$

Where; $PayGap \times MSH_{it}$ is the interaction term, PayGap1 × MSH and PayGap2 × MSH, alternatively. The rest of the variables are the same as in equation (3). Table 4, columns 2–3, provides intriguing insights regarding the moderating role of CEO shareholding (MSH) in the nexus. The results reveal a significant and negative direct effect of M_SH on digital transformation. Furthermore, the interaction terms PayGap1 × M_SH and Paygap2 × M_SH in relation to digital transformation have positive coefficients (with coefficients of 0.048*** (SE = 0.011) and 0.050*** (0.012), respectively). It shows that Managerial Shareholding positively moderates the CEOs pay gap and digital transformation relationship. These findings substantiate hypothesis (H2) that managerial shareholding helps align the interests of CEOs with other stakeholders, thus enhancing the positive impact of CEOs pay gap on digital transformation.

5.4. CEO pay gap and digital transformation nexus: the effect of SOEs vs. non-SOEs

Different ownership structures, especially in the context of China, where state ownership plays a significant role, can impact the top management incentives and policies (Jin et al., 2022). Therefore, the study investigates the impact of State-Owned Enterprises (SOEs) vs non-State-Owned Enterprises (non-SOEs) on the association between CEOs pay gaps and DT. To analyze this, two interaction terms (PayGap1 × SOE) and (PayGap2 × SOE) and estimated the model as explained in the following Equation (5) under hypothesis 3 (H3).

Table 4
Moderating effect of managerial shareholding and ownership structure (SOEs vs non-SOEs) in CEOs pay gap and DT nexus.

DV = DT	Managerial Shareholding		Ownership structure (SOEs vs non-SOEs)	
	(1)	(2)	(3)	(4)
PayGap1	0.066***		0.132***	
	(0.009)		(0.008)	
PayGap2		0.087***		0.157***
		(0.011)		(0.009)
MGSH	−0.603***	−0.663***		
	(0.153)	(0.179)		
PayGap1 × MGSH	0.048***			
	(0.011)			
PayGap2 × MGSH		0.050***		
		(0.012)		
SOE			0.762***	1.152***
			(0.149)	(0.170)
PayGap1 × SOE			−0.069***	
			(0.011)	
PayGap2 × SOE				−0.092***
				(0.012)
Size	0.006	0.002	0.002	−0.001
	(0.005)	(0.005)	(0.005)	(0.005)
Lev	−0.107***	−0.094***	−0.082***	−0.071**
	(0.030)	(0.029)	(0.030)	(0.029)
Age	−0.097***	−0.094***	−0.069***	−0.068***
	(0.010)	(0.009)	(0.010)	(0.009)
ROA	0.049	0.066	0.104	0.119
	(0.084)	(0.080)	(0.084)	(0.080)
CR	−0.142***	−0.147***	−0.137***	−0.142***
	(0.052)	(0.050)	(0.052)	(0.050)
PPE	−0.245***	−0.230***	−0.222***	−0.210***
	(0.044)	(0.043)	(0.044)	(0.043)
BoSize	−0.027***	−0.031***	−0.016***	−0.020***
	(0.004)	(0.004)	(0.004)	(0.004)
IndDir	0.003***	0.004***	0.005***	0.006***
	(0.001)	(0.001)	(0.001)	(0.001)
N_employee	0.046***	0.040***	0.050***	0.046***
	(0.005)	(0.005)	(0.005)	(0.005)
FCEO	−0.068**	−0.063*	−0.062*	−0.061*
	(0.034)	(0.033)	(0.034)	(0.033)
Duality	0.129***	0.131***	0.091***	0.102***
	(0.014)	(0.013)	(0.014)	(0.013)
Constant	−0.289	−0.508***	−1.210***	−1.559***
	(0.180)	(0.192)	(0.164)	(0.169)
Industry FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Observations	32332	32332	32332	32332
R-squared	0.172	0.175	0.176	0.178

Notes: Standard errors are in parenthesis ***p < 0.01, **p < 0.05, *p < 0.1. Variables definitions are given in Table A1(Appendix). Estimated Models are explained in section 5.3 and 5.4, equations (4) and (5), respectively.

$$DT_{it} = \alpha + \beta_1 PayGap_{it} + \beta_2 PayGap \times SOE_{it} + \beta_n control_{it} + year_t + industry_i + \varepsilon_{it} \quad (5)$$

Where; $PayGap \times SOE_{it}$ is the interaction term, PayGap1 × SOE and PayGap2 × SOE, alternatively. The rest of the variables are the same as in equation (3). This approach allows for a comparative analysis of how the CEO pay gap influences digital transformation in state-owned versus privately-owned firms. The results reported in Table-4, columns 4–5, show that the coefficients of Paygap1 and Paygap2 are significant positive for direct effect of SOEs on DT. Moreover, the interaction terms PayGap1 × SOE and Paygap2 × SOE in relation to digital transformation have significant negative coefficients (with coefficients of −0.069*** (S.E. = 0.011) and −0.092*** (S.E. = 0.012), respectively). It shows that SOEs negatively moderate the CEOs pay gap and digital transformation relationship. These findings substantiate hypothesis (H3).

5.5. Effect of government CEOs pay regulations

In China, state-owned businesses have transformed into the majority of publicly traded companies. State-owned firms prioritize equitable compensation, while authorities enforce restrictions on the disparity in remuneration between senior executives and workers (Yin & Du, 2021a, b). The Chinese government has implemented policies to control excessive executive pay, particularly in state-owned enterprises (SOEs), to narrow the pay gap between executives and average employees. The Chinese government initiated these policies in 2009 and strictly enforced them in 2014. This approach is designed to promote social equity and mitigate public discontent, as noted by Maqsood, Li, et al. (2024). By capping CEO compensation, the government seeks to prevent excessive risk-taking and ensure that executive incentives align with national economic goals. Such restrictions significantly influence the relationship between CEO compensation and digital transformation (DT). The primary aim of these regulations is to address income disparities and ensure that executive compensation aligns with broader economic and social objectives. These regulatory frameworks can limit the flexibility of SOEs in designing performance-based compensation packages. With restrictions on pay gaps, the potential for offering substantial financial incentives linked to DT initiatives is constrained, potentially reducing the motivation for CEOs to aggressively pursue DT, as highlighted by Jiang and Zhang (2018). Therefore, Chinese government regulations on CEO pay gaps create a complex impact on the relationship between CEO compensation and digital transformation. By limiting excessive financial incentives, they may constrain the ability of enterprises, particularly SOEs, to use compensation as a direct motivator for aggressive DT efforts.

To analyze the impact of government regulation on CEO pay and its effect on enterprises' digital transformation (DT), a Difference-in-Difference (DiD) regression model is estimated as given in the following equation (6):

$$DT_{it} = \alpha + \beta_1 Post_{it} + \beta_2 SOEs_{it} + \beta_3 \text{SOEs} \times \text{Post}_{it} + \beta_n control_{it} + year_t + industry_i + \varepsilon_{it} \quad (6)$$

Here $Post_{it}$ is the policy variable having the value 1 for the years 2014 and later; 0 otherwise. The rest of the variables are same as in equation (3). The results for the treated group (SOEs), shown in Table 5, columns 2 and 3, reveal coefficients of 0.217*** (S.E. = 0.024) and 0.152*** (S.E. = 0.025), respectively. These values suggest that State-Owned Enterprises (SOEs) are more inclined to incorporate digital transformation (DT) into their operations. The interaction term **SOEs × Post**, which captures the effect of being an SOE after the implementation of pay regulation, shows a negative impact of −0.327*** (S.E. = 0.014) in column 1 when control variables are not included. Even after accounting for other factors by adding control variables in column 2, the coefficient remains negative at −0.300***(S.E. = 0.014). These results highlight the substantial negative consequences of government regulation of CEO compensation on DT. In simpler terms, while the regulation aims to create a fairer pay structure, it might inadvertently slow down the adoption of digital technologies within SOEs. This comprehensive analysis underscores the importance of carefully considering the design and implementation of CEO compensation regulations. Policymakers need to be aware that such regulations can have wide-ranging effects on firm performance metrics and technological advancements, which might not always be positive.

Table 5
Government CEO pay regulations and DT.

	DV = DT	DV = DT
	(1)	(2)
Post	1.215***	1.128***
	(0.021)	(0.022)
SOEs	0.217***	0.152***
	(0.024)	(0.025)
SOEs × Post	−0.327***	−0.300***
	(0.014)	(0.014)
Constant	0.703***	−0.666***
	(0.046)	(0.128)
Controls	YES	YES
Industry FE	YES	YES
Year FE	YES	YES
Observations	32588	32028
R-squared	0.302	0.314

Notes: Standard errors are in parenthesis ***p < 0.01, **p < 0.05, *p < 0.1. Variables definitions are given in Table A1(Appendix). Control variables are included but the results are not reported for brevity. Estimated Model is explained in section 5.5, equation (6).

5.6. Endogeneity and robustness test

To control for confounding variables, enhancing the robustness and reliability of the findings, and ensuring the validity of the findings, following tests such as alternative sample and variable measures, two-stage least squares (2SLS) with instrumental variables, and Propensity Score Matching (PSM) were employed.

5.6.1. Alternative measures of DT and CEOs pay gap

In the sensitivity analysis, the regression model (3) is re-estimated with an alternative proxy of CEO paygap and DT. The CEO paygap is replaced with PayGap3 (the difference between the average pay of CEOs and employees), and the results are shown in column 4 of Table 6, and DT is replaced by DT2. The DT2 represents the total frequency of the keywords; the results are shown in columns 2 and 3 of Table 6. The link between CEO pay and DT remains positive and statistically significant after these substitutions, confirming that the findings are robust and free from bias.

5.6.2. Two-stage least squares (2SLS)

To address potential endogeneity, building on the work of Cai et al. (2021), the study computes an instrumental variable by taking the natural logarithm of the pay disparity between CEOs and the average pay of executives. This instrument is used in a two-stage least squares (2SLS) estimation because it is strongly correlated with the CEO pay gap but is unlikely to directly affect the dependent variables. The Durbin score and the Wu-Hausman test are used to assess the validity of instrumental variables. Results reported at the bottom of Table 6 indicates that the instrument variables are extremely endogenous as the

Table 6
Alternative measures of CEOs pay gap and DT.

	DV = DT2		DV = DT
	(1)	(2)	(3)
PayGap1	0.568***		
	(0.146)		
PayGap2		0.393**	
		(0.178)	
PayGap3			0.045***
			(0.009)
Constant	−34.141***	−29.102***	−1.146***
	(3.953)	(3.933)	(0.165)
Controls	YES	YES	YES
Industry FE	YES	YES	YES
Year FE	YES	YES	YES
Observations	32332	32332	32468
R-squared	0.132	0.128	0.307

Notes: Standard errors are in parenthesis ***p < 0.01, **p < 0.05, *p < 0.1. Variables definitions are given in Table A1(Appendix). Control variables are included but the results are not reported for brevity. Estimated Model is explained in section 4.5, equation (1) with replacing the CEOs pay gap and Digital transformation variables.

both tests scores are insignificant with the values (5.533, 0.274, 5.529 and 0.271, respectively). The results of the 2SLS regression are shown in Table 6, Panel A. Columns 2–3 present the first stage results, while columns 4–5 show the second stage. The coefficients for CEO Paygap1 and Paygap2 are positive and statistically significant for DT, suggesting that Hypothesis 1 (H1) holds even after addressing endogeneity concerns.

5.6.3. *Propensity score matching (PSM)*

To address self-selection bias, the study re-estimates model 3 using nearest-neighbor (NN) matching within the framework of propensity score matching (PSM). The primary objective of using PSM is to balance observable covariates, such as Size, Leverage, Age, ROA, Cash Ratio, PPE, Board Size, IndDir, number of employees, FCEO and Duality, between the treatment group (Pay gap above median = 1) and the control group (Pay gap above median = 0). Including these control variables ensures that the distribution of treatment and control participants is balanced in the matched sample, which is a critical criterion for PSM (Austin, 2011). If there are no systematic changes between these two groups in the baseline variables, after conditioning on the propensity score, the propensity-scoring model has been adequately defined (Austin, 2011).

The PSM matched sample results, shown in Table 7, Panel B (columns 6–7), validate the primary findings. All coefficients related to the CEO's pay gap are positive and statistically significant. Fig. 3 illustrates the balancing test, which reveals no substantial differences in the mean values of variables before and after matching. Fig. 4 further show that the matching effect passes the common support test, as the kernel density function graphs are nearly identical pre- and post-matching. These results confirm that the study effectively addresses self-selection bias.

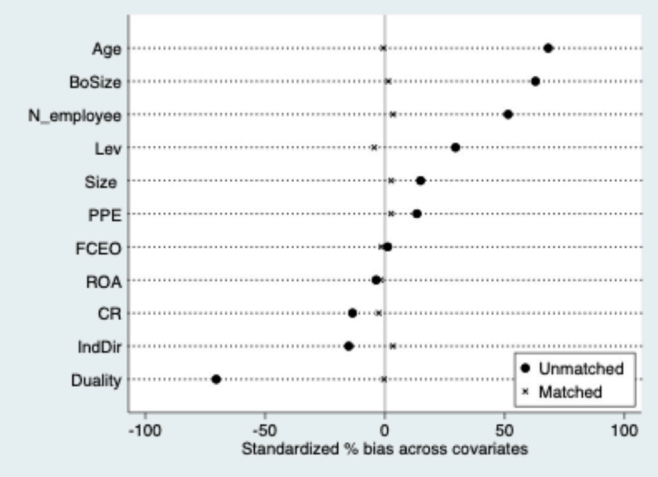


Fig. 3. Balancing test.

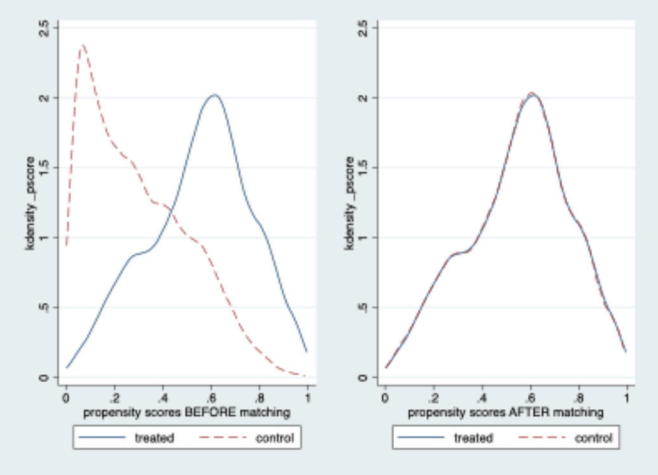


Fig. 4. Represents the line of propensity score before and after the matching.

5.7. *Heterogeneous analysis*

To examine the heterogeneity of Paygap's role in enhancing DT adoption, this study classified the sample into subsamples based on different regions and time periods. To begin with Covid-19 period, there was rapid shift in the adoption of DT during and after the Covid-19 outbreak led the study to divide the overall sample into two subperiods: Pre-Covid (2011–2020) and post-Covid (2021–2023). Table 8, Panel A: columns 2 & 3 show a significant positive relationship between paygap and DT during the pre-COVID period. However, a negative but insignificant relationship emerged during the post-COVID period. The

Table 7
Robustness analysis (2SLS and PSM regression) results.

VARIABLES	Panel A: 2SLS				Panel B: PSM	
	first		Second			
	(1)	(2)	(3)	(4)	(5)	(6)
	Paygap1	PayGap2	DT	DT	DT	DT
IV	1.191***	1.107***				
	(0.005)	(0.027)				
PayGap1			0.067***		0.036***	
			(0.008)		(0.006)	
PayGap2				0.066***		0.050***
				(0.008)		(0.008)
Constant	−4.714***	−2.479***	19.649***	−0.094	−0.577***	−0.813***
	(0.078)	(0.041)	(0.126)	(0.230)	(0.163)	(0.166)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Durbin (score)			5.533	0.274		
Wu-Hausman			5.529	0.271		
Observations	30,060	30,060	30,060	30,060	17723	18866
R-squared	0.721	0.883	0.239	0.241	0.295	0.295

Note: Standard errors are in parenthesis ***p < 0.01, **p < 0.05, *p < 0.1 Instrument Variable (2sls regression) = Paygap3. Control variables are included but the results are not reported for brevity.

Table 8
Heterogeneous analysis and reduced sample.

DV = DT3	(1)	(2)	(3)	(4)
Panel A: Reduced Sample	Pre Covid (2011–2020)		Post Covid (2021–2023)	
PayGap1	0.020***		−0.016	
	(0.007)		(0.011)	
PayGap2		0.032***		−0.020
		(0.008)		(0.014)
Constant	−0.449**	−0.514***	0.271	0.296
	(0.176)	(0.175)	(0.421)	(0.419)
Controls	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	20714	22308	9252	10024
Within R^2	0.267	0.265	0.059	0.056
Panel B: Fintech region	**YES**		**No**	
PayGap1	0.009*		0.007	
	(0.009)		(0.009)	
PayGap2		0.045***		0.015
		(0.011)		(0.011)
Constant	−0.331	−0.749**	−0.449	−0.551*
	(0.308)	(0.306)	(0.289)	(0.288)
Controls	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	12832	13820	10147	10947
Within R^2	0.303	0.297	0.274	0.280
Panel C: Bank competition	**High**		**Low**	
PayGap1	0.001		0.027***	
	(0.008)		(0.008)	
PayGap2		0.017		0.034***
		(0.010)		(0.010)
Constant	−0.692***	−0.770***	−0.335	−0.507**
	(0.266)	(0.262)	(0.231)	(0.233)
Controls	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	13723	14810	13072	14103
Within R^2	0.303	0.299	0.282	0.282

Note: Standard errors are in parenthesis ***p < 0.01, **p < 0.05, *p < 0.1. Control variables are included but the results are not reported for brevity.

shift in priorities during the immediate aftermath of the COVID-19 pandemic saw Chinese companies focus on operational stability and cost-cutting over new digital technology investments, shifting from long-term strategies like DT to short-term survival tactics and reducing the impact of CEO pay gaps on DT efforts. The rapid and widespread adoption of digital technologies during the pandemic may have leveled the playing field, making DT a necessity rather than a strategic choice and diminishing the relevance of CEO incentives (Hou et al., 2024). Post-COVID, increased regulatory scrutiny and changes in market dynamics, including policies promoting fairness and equity in pay, likely diluted the incentive effects of pay gaps on DT (Li et al., 2024). The focus on crisis management and resilience may have overshadowed DT initiatives, with CEOs directing efforts towards navigating the immediate impacts of the pandemic (Huang & Zhao, 2024). Furthermore, the crisis may have fostered a culture of increased collaboration and shared leadership, reducing the influence of individual CEO incentives on DT.

To account for the heterogeneous effects of regional resource availability on DT, the study considers the varying levels of support for DT in different regions. Firms located in well-developed fintech regions, such as Beijing, Guangzhou, Shanghai, and Shenzhen, benefit from abundant resources that facilitate DT. A subsample analysis is conducted, comparing firms in high-fintech regions (e.g., Beijing, Guangzhou, Shanghai, and Shenzhen) to those in low-tech regions, following the approach of Zhai et al. (2022). The results, presented in Table 8, Panel B (columns 2–3), indicate that the coefficients for PayGap1 and PayGap2 are significantly positive for firms in high-fintech regions. This suggests that in regions with robust financial technology sectors, larger CEO pay gaps continue to incentivize greater engagement in DT initiatives.

Panel C examines the influence of external factors, specifically bank concentration, on the relationship between CEO pay gaps and digital transformation (DT). The analysis shows that in regions with low bank competition, the impact of PayGap on DT is significantly positive. This suggests that in areas where a few banks dominate the market, firms may face more stringent financing conditions, making internal incentives like CEO pay gaps more crucial for driving DT initiatives. Conversely, in regions with high bank concentration, the positive impact of PayGap on DT is less pronounced, indicating that the competitive banking environment influences the effectiveness of CEO incentives in promoting digital transformation efforts.

5.8. Influencing mechanism analysis

The relationship between CEO pay gaps and digital transformation (DT) can be examined through the lens of agency theory, which posits that aligning the interests of executives with those of shareholders helps reduce agency costs and improves organizational performance. In this context, a well-structured CEO pay gap can facilitate DT by lowering agency costs and improving audit quality.

5.8.1. Agency cost: increase in governance quality

The CEO pay gap, when aligned with long-term strategic goals, can decrease agency costs by ensuring that executives are motivated to act in the best interests of the company and its shareholders. By linking a significant portion of CEO compensation to performance metrics related to DT, companies can incentivize executives to focus on initiatives that drive innovation and long-term value creation. This alignment reduces the likelihood of executives pursuing personal agendas that do not align with shareholder interests, thereby decreasing agency costs (Rashid, 2016). When executives are financially motivated to prioritize DT, they are more likely to make decisions that support the company's digital strategy, ensuring that resources are allocated efficiently and effectively (Hou et al., 2024). So, in the path/channel analysis of CEOs, pay gap and digital transformation to reduce the agency problem, the agency cost (AgenEXP) is used as the intermediary variable. Following equations (7) and (8) explaining the channel analysis, in the first step, CEOs pay gap is regressed on the agency cost as explained in equation (5). Then in the second step, both the CEOs pay gap and agency costs are regressed on the digital transformation.

$$\mathrm{AgenEXP}_{i,t} = \alpha + \beta_1 Paygap_{i,t} + \beta_n control_{i,t} + year_t + industry_i + \varepsilon_{i,t} \quad (7)$$

$$DT_{i,t} = \alpha + \beta_1 Paygap_{i,t} + \beta_2 \mathrm{AgenEXP}_{i,t} + \beta_n control_{i,t} + year_t + industry_i + \varepsilon_{it} \quad (8)$$

Where: $\mathrm{AgenEXP}_{i,t}$ is the agency cost and rest of the variables are the same as in equation (3).

The results of channel analysis, i.e., equations (7) and (8), are then reported in Columns 2–5 of Table 9, Panel A. Models (1) and (2) report the regression results of the paygap effect on agency cost, and models (3) and (4) display the results of the mediation effect of agency cost in the paygap and digital transformation relationship. The findings reveal significant negative results regarding the relationship between the CEO pay gap and corporate agency costs. For AgenEXP, both Paygap1 and Paygap2 show strong negative correlations, with coefficients of −0.284*** and −0.360***, respectively, at the 1 % significance level. Additionally, the regression coefficients for Paygap1, Paygap2, and AgenEXP on digital transformation (DT) are 0.083*** and 0.102***, respectively, both significant at the 1 % level. Thus, the analysis confirms a mediating effect: agency costs negatively mediate the impact of the pay gap on DT. In other words, a larger CEO pay gap reduces agency

Table 9
Mediating role of agency cost and audit quality in CEO PayGap and DT nexus.

Panel A: Agency cost	AgenEXP		DT3	
	(1)	(2)	(3)	(4)
PayGap1	−0.284***		0.083***	
	(0.007)		(0.006)	
PayGap2		−0.360***		0.102***
		(0.008)		(0.007)
AgenEXP			−0.044***	−0.045***
			(0.005)	(0.005)
Constant	−6.111***	−5.110***	−0.936***	−1.153***
	(0.167)	(0.168)	(0.153)	(0.153)
Controls	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	29825	32189	29754	32108
R-squared	0.580	0.589	0.172	0.175
Audit quality	**Big4**		**DT3**	
PayGap1	0.195***		0.046***	
	(0.003)		(0.006)	
PayGap2		0.240***		0.060***
		(0.004)		(0.007)
Big4			0.253***	0.247***
			(0.012)	(0.011)
Constant	8.860***	8.200***	−2.897***	−2.943***
	(0.075)	(0.077)	(0.181)	(0.174)
Controls	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	29864	32221	29792	32139
R-squared	0.862	0.863	0.183	0.186

Notes: Standard errors are in parenthesis ***p < 0.01, **p < 0.05, *p < 0.1. Variables definitions are given in Table A1(Appendix). Panel A, Estimated Model is explained in section 5.7.1, equations (7) and (8). Panel B, Estimated Model is explained in section 5.7.2, equations (9) and (10).

costs, thereby promoting digital transformation.

5.8.2. Audit quality: decrease in agency issues

Secondly, enhancing audit quality is another mechanism through which CEO pay gaps can promote digital transformation (DT). High audit quality ensures that a company's financial statements accurately reflect its financial position and performance (Zahid et al., 2024). When CEO compensation is tied to transparent and rigorous performance metrics, it necessitates strong auditing processes to ensure accurate reporting and evaluation. This focus on audit quality leads to more reliable financial reporting, which supports informed decision-making and builds trust with investors and other stakeholders, making it easier for companies to secure the necessary investment (Zahid, Taran, et al., 2023) and support for DT initiatives.

Audit quality mediates the positive relationship between the CEO pay gap and DT. As Big4 auditors ensure the quality of financial statements, they serve as a proxy for financial reporting quality and the overall information environment. This increases stakeholder confidence in the firm's management, aligning the interests of CEOs with those of stakeholders and reducing agency issues, further facilitating digital transformation. Following equations (9) and (10) explaining the channel analysis, in the first step, CEOs pay gap is regressed on the big 4 auditors as explained in equation (9). Then in the second step, both CEOs pay gap and big 4 auditors are regressed on the digital transformation.

$$AuditbyBig4_{i,t} = \alpha + \beta_1 Paygap_{i,t} + \beta_n control_{i,t} + year_t + industry_i + \varepsilon_{i,t} \quad (9)$$

$$DT_{i,t} = \alpha + \beta_1 Paygap_{i,t} + \beta_2 aduitbyBig4_{i,t} + \beta_n control_{i,t} + year_t + industry_i + \varepsilon_{it} \quad (10)$$

Where: $AuditbyBig4_{i,t}$ is the audit quality and rest of the variables are the same as in equation (3). The results of equations (9) and (10) are presented in columns 2–5 of Table 9, Panel B.

Models (1) and (2) report the regression results of the paygap effect on audit quality, and models (3) and (4) display the results of the mediation effect of audit quality in the paygap and digital transformation relationship. The findings indicate that Paygap1 and Paygap2 are positively correlated with audit quality (proxied by Big4), with coefficients of 0.195 and 0.240, respectively, both significant at the 1 % level, demonstrating a significant positive relationship between the CEO pay gap and audit quality. The regression coefficients for Paygap1 and Big4 on DT are 0.046 and 0.60, respectively, both significant at the 1 % level. Similarly, the coefficients for Paygap2 and Big4 on DT are 0.065 and 0.422, respectively, also significant at the 1 % level. These results remain robust even after including control variables, confirming the mediating effect of audit quality in the relationship between the CEO pay gap and digital transformation.

6. Conclusions

This digital revolution profoundly impacts how businesses operate. To remain successful and competitive, businesses must adapt to these dynamics and drive digitalization initiatives. Executives shape digital strategies, yet motivations behind CEO-led digitalization lack academic discussion. This study has provided significant insights into the relationship between CEO compensation structures and digital transformation (DT) within firms, with a particular focus on Chinese enterprises. By examining the nuanced mechanisms, contextual factors, and temporal dynamics that influence this relationship, the research has addressed key gaps in the literature and advanced the understanding of how executive incentives can drive or hinder DT efforts. Empirical findings show that a pay gap between CEOs and employees motivates investment in digital technologies and embraces digital transformation. Moreover, the positive relationship between CEOs pay gap and digital transformation is more pronounced in SOEs than in non-SOEs. Additionally, shareholding by the CEOs of the company strengthens this relationship. These findings align with the agency theory, suggesting that a high CEO pay gap incentivizes CEOs to align their interests with the digital transformation goals of the organization. The results remain consistent across different measures and unaffected by endogeneity problems.

This study highlights the importance of CEO compensation incentives in driving digital transformation (DT) within firms, benefiting shareholders by aligning managerial and investor interests. Effective incentive reform is crucial for successful long-term digital initiatives, extending agency theory in the context of executive incentives and innovation. Additionally, the study demonstrates the moderating role of contextual factors, such as ownership structure and managerial shareholding, in this relationship. Findings show that aligning executive compensation with digital performance metrics can mitigate agency costs and enhance strategic decision-making. Shareholder use of management shareholding can further align CEO interests, emphasizing the need to consider ownership types, particularly in economies like China with significant state-owned enterprises (SOEs). The study contributes to the agency theory by demonstrating how CEO compensation can mitigate agency problems and promote DT, exploring variations based on CEO shareholding and firm ownership structure to provide insights into how compensation policies impact digital initiatives. For practitioners, this study underscores the critical importance of designing effective executive compensation packages to foster DT. The findings suggest that firms should carefully structure CEO incentives to align with long-term digital strategies.

However, the study's generalizability is limited by sample specificity and time constraints, despite efforts to control for industry and year variations. Factors like corporate culture and leadership style may also influence the CEO pay gap and digital transformation relationship. While the panel analysis offers insights, it doesn't establish strict causation. Future studies could benefit from case studies that generalize

results. Lastly, focusing on Chinese companies may limit extrapolation to other regions. Future research should explore this relationship across different remuneration types, industries, and regions, considering moderating factors.

CRediT authorship contribution statement

R.M. Ammar Zahid: Writing – original draft, Conceptualization. **Umer Sahil Maqsood:** Writing – review & editing, Methodology. **Murtaza Hussain:** Visualization, Software. **Muhammad Waleed Younas:** Data curation.

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Appendix A. Supplementary data

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Data availability

Data will be made available on request.

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